Chapter 12: Architecture Development Method

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 12 (pp. 295-560)

Overview

Chapter 12 (pp. 295-560) is the longest chapter in the book. It walks through the **Architecture Development Method (ADM)**, "the heart of the TOGAF Standard" (p. 295). Section 12.1 introduces the ten phases and their objectives (pp. 296-300). Section 12.2 explains the four key documents created along the way: Request for Architecture Work, Statement of Architecture Work, Architecture Definition Document, and Architecture Contract (pp. 300-305). Sections 12.3 to 12.12 cover each phase in turn: its steps (in tables), Jager's short "steps" boxes, and the artifacts it can produce, all illustrated with the fictional healthcare organization. The phases are Preliminary (12.3), A Architecture Vision (12.4), B Business (12.5), C Information Systems, both Data and Application (12.6), D Technology (12.7), E Opportunities and Solutions with the Architecture Roadmap (12.8), F Migration Planning (12.9), G Implementation Governance with performance monitoring (12.10), H Architecture Change Management (12.11), and Requirements Management (12.12). Section 12.13 assembles the ADM documents, and Section 12.14 summarizes. Leave with this idea: the ADM describes everything you *can* do, not what you *must* do. It does not prescribe an order and must be tailored. It is not a waterfall (p. 295). Chapter 12 is part of the PINK, GREEN, and BLUE reading tours (p. 295).

> **Reading note:** The phase objectives, steps, and artifact names follow the Standard. The healthcare examples, the shaded "steps" boxes, and the personal stories (hospital strategy documents, p. 309; others later) are Jager's. Jager also adds artifacts the Standard lacks, such as the Work Package Portfolio Map (see Ch. 6 and 14). Each phase has the same fixed components: **key objectives, steps, outputs, and approach** (p. 300).

Detailed Section Breakdown

Chapter Opener (p. 295)

- **Iterative, not waterfall.** The ADM guides the creation and maintenance of artifacts aligned with business goals. "Don't be fooled by what seems to be a waterfall approach at first sight" (p. 295).
- **Descriptive, not prescriptive.** It "describes all the actions one can take, not what must be taken", does not prescribe an order, and is to be tailored (p. 295).

12.1 The Heart of the TOGAF Standard (pp. 296-300)

- **A method, not a process.** It is "still too often mistaken for a process". Its steps are possible steps that may be skipped or adapted (p. 296).

Phase	Key objective(s)	Page
Preliminary	Determine the Architecture Capability the organization needs, and establish it	p. 297
A: Architecture Vision	A high-level vision and stakeholder alignment, with business requirements articulated and agreed	p. 298

Phase	Key objective(s)	Page
B, C, D: Business, Information Systems, Technology	(1) Craft the Target Architecture. (2) Identify candidate Architecture Roadmap components from Baseline-Target gaps	p. 298
E: Opportunities and Solutions	Identify work packages; decide whether an incremental approach is needed; define the building blocks to complete the target	p. 298
F: Migration Planning	A detailed implementation plan	p. 298
G: Implementation Governance	Ensure adherence, manage changes during implementation, assess progress	p. 299
H: Architecture Change Management	Continuously assess changes in strategy, environment, and technology that affect the architecture	p. 299
Requirements Management	Not a separate phase: a continuous activity managing requirements throughout	p. 299

- **Artifacts must add value.** "A picture is worth a thousand words". The chapter's artifacts are healthcare examples to adapt (pp. 299-300).

12.2 Architecture Document Creation (pp. 300-305)

Document	Purpose	Created / approved	Page
Request for Architecture Work	Formal, high-level request from the sponsoring organization to the architecture team that triggers a cycle	Preliminary Phase	pp. 300, 302
Statement of Architecture Work	Scope and approach of the cycle. A benchmark for success and effectively a contract between the architecting organization and the sponsor (project description and scope, vision overview, acceptance criteria, plan, approvals)	Phase A	pp. 301, 303
Architecture Definition Document	"Deliverable container" for core artifacts across all four domains and all states (Baseline, Transition, Target)	Built in B-D. Target approved in E (with draft Transitions). Finalized in F	pp. 301, 303-304
Architecture Contract	Joint agreement between development partners and sponsors on deliverables, quality, and fitness for purpose	Phase E onward. Starts transition and governance	pp. 301, 304-305

- **Flow (Figure 12-2).** Request (Preliminary). Statement with candidate building blocks (A). Baseline then Target descriptions per domain, reusing the repository (B-D). Building blocks recorded in the Architecture Landscape, future standards in the repository. Gap analysis then Architecture Roadmap. Stakeholder reviews. ADD approved (E) and finalized (F). Missing blocks linked to work packages (E). Chosen solution in an Architecture Contract (pp. 302-304).
- **Loop back.** If change during execution needs major architecture changes, a new Request for Architecture Work starts a new ADM iteration (p. 305).

12.3 Preliminary Phase (pp. 305-323)

- **Purpose.** Get the organization ready: align stakeholders, define principles, tailor TOGAF, and put frameworks, tools, and governance in place. Skipping it risks misalignment and poor management of the initiative (p. 305).

12.3.1 Determining and Establishing the Architecture Capability (pp. 305-319)

- **Two objectives.** *Determine* the capability needed (scope, stakeholders, their requirements, current and target maturity, readiness for change) and *establish* it (pp. 305-306).

Step (Table 12-1, determining)	Description	Output	Page
Define the enterprise	Scope the impacted organization and its structure. Identify stakeholders	Scope of impacted organization(s)	pp. 306-307
Identify key drivers and organizational context	Find the factors that influence the architecture	Baseline of objectives, goals, drivers	pp. 306, 308
Manage relationships between frameworks	Understand how management frameworks interact. Set up a framework to manage architecture activities	EA Capability structure. Architecture Governance Framework	pp. 306, 310-312
Assess EA maturity	Evaluate where the organization stands	Maturity assessments, gaps, strategies	pp. 306, 312-313

- **Define the enterprise.** Stakeholders include those most affected who gain most value, those whose work changes, and external parties. Meet them, discuss the scope, recruit more stakeholders, and iterate (Stakeholder Management, Ch. 14.3) (p. 307).
- **Key drivers.** No earlier phase gives input, so use the TOGAF Library, other frameworks, and business and IT strategies, principles, goals, and drivers. Distill them from documents and verify them with stakeholders (p. 308).
- **Manage relationships between frameworks.** Reuse existing maturity assessments, roles, budgets, methods, tools, and principles, but remove what is outdated rather than copying it. Use PPM frameworks too. Set up an Architecture Governance Framework covering how architecture materials are managed, repository requirements, and governance processes (compliance, retirement), with stakeholder reviews of the models (pp. 310-312).
- **Assess maturity.** The "Architecture Maturity Models" Series Guide names ACMM (often IT-related) and CMMI (originally software and engineering, now also used for EA because of its business-goal focus). Choose what fits (Ch. 15) (pp. 312-313).
- **Readiness for change.** Use the Business Transformation Readiness Assessment (Ch. 14.7) (p. 313).

Step (Table 12-2, establishing)	Description	Output	Page
Establish the EA team	Form a team to do the architecture work	Architecture team. Defined roles and responsibilities	pp. 314-316
Set Architecture Principles	Define principles guiding all architecture efforts	Architecture Principles. Constraints for architecture efforts	pp. 314, 316-317
Choose and tailor the architecture framework	Tailor the TOGAF Standard, the ADM, and the Content Framework	Tailored framework, ADM, Content Framework	pp. 314, 317-318

- **The team.** Not only architects: project managers, senior managers, and others who grasp architecture can join. Composition follows positioning (IT-based means architects plus technical experts; strategic means EA plus senior and portfolio managers). "Architecture teams" are not the same as teams of people with the job title architect (pp. 314-315).
- **Example RACI (Table 12-3).** The Enterprise Architect is Accountable for every activity and also Responsible for creating the vision and implementing governance. Segment Architects are Responsible for Segment Architectures, opportunities and solutions, the migration plan, change, and requirements. Solution Architects are Responsible for opportunities, migration, change, and requirements, and Consulted on the rest (pp. 315-316).

- **Tailoring sequence.** Agree on and publish a glossary. Adapt the Enterprise Metamodel (new entities, deliverables, artifacts). Adapt the ADM (remove redundant tasks, add checkpoints, integrate with other processes). Customize the Content Framework. Set a tools strategy (general and specialist tools, business modelling). Formalize change control (pp. 317-318).
- **Not isolated.** EA frameworks must interoperate with broader business frameworks. Partition teams, processes, and deliverables to fit size and budget. You may need to revisit the Preliminary Phase from Phase A (p. 319).
- **In essence.** "The Preliminary Phase is about building the architecture needed to support the Architecture Capability" (p. 319).

Jager's commentary (author's view): In a hospital with little EA, he hand-mined stacks of strategy documents for drivers, goals, and objectives, built a Driver/Goal/Objective Diagram (Figure 12-3), and refined it with senior management into a Solution Concept Diagram. Today AI tools can speed up this scanning (pp. 309-310). Architecture decisions belong to stakeholders, not the architect (p. 311).

12.3.2 Preliminary Phase Artifacts (pp. 320-323)

- **Principles Catalog** (entity: principle). It holds Enterprise Principles (high-level guidelines shaping strategy, culture, decisions) and Architecture Principles (what a good solution looks like). Principles are used to decide at decision points and to govern change (p. 320).
- **Example principles (Table 12-5):** Comply with laws and regulations; Unambiguous ways of working; Store information once and use it more often (data sharing will "bump up against" data security); Standard products and services (no duplicative capabilities). Each has a Name, Statement, Rationale, and Implications (pp. 321-323).

12.4 Phase A: Architecture Vision (pp. 324-347)

- **The first actionable phase.** It sets the objectives, scope, and expected outcomes, building on Preliminary input (p. 324).

12.4.1 Creating the Architecture Vision (pp. 324-335)

- **Main goal.** A high-level vision and stakeholder alignment, with clearly agreed business requirements and initial approval to proceed. The result goes in an architecture vision document (p. 324).
- **Inputs.** Preliminary Phase reference material, business principles, goals, and drivers. Existing documentation is used to populate the first version of the Architecture Repository, e.g., starting capabilities as ABBs in the Architecture Landscape and the RACI in the Architecture Capability section (p. 324).

Step (Table 12-6)	Description	Output	Page
Establish the architecture project	Formal project recognition, leadership endorsement, line management support	Statement of Architecture Work	pp. 325-327
Identify stakeholders and determine requirements	Identify stakeholders and requirements so the work can be communicated and aligned	Stakeholder Catalog; Communications Plan	pp. 325, 327-328
Assess readiness for business transformation	Evaluate change readiness to guide scope and flag attention areas	Business Transformation Readiness Assessment	pp. 325, 329-330
Elaborate on business goals and Architecture Principles	Validate and refine goals and principles from the Preliminary Phase	Validated objectives, goals, drivers, principles	pp. 325, 330-331
Create a high-level architecture vision	High-level solution description aligned with strategy	Solution Concept Diagram	pp. 325, 331-332
Define the Target Architecture value propositions and KPIs	Business case, performance metrics, and measures	Value propositions and KPIs	pp. 325, 333

Step (Table 12-6)	Description	Output	Page
Identify business transformation risks and mitigation activities	Identify risks, assess their level and frequency	Risks; mitigating activities	pp. 326, 333-334
Develop Statement of Architecture Work	Draft it and secure formal approval	Statement of Architecture Work	pp. 326, 334-335

- **Run it as a project.** Each ADM iteration is ideally a project in the existing project management framework, standalone or part of broader programs (p. 326).
- **Stakeholders continue from Preliminary.** Use stakeholder analysis, mapping, and a Communications Plan (Ch. 14.3.1-14.3.3). Stakeholder requirements and concerns feed the Solution Concept Diagram (pp. 327-328).
- **Readiness factors (examples).** Vision, leadership (executive sponsorship), stakeholder engagement, governance, culture and communications. Assess each for risks and interventions. The results shape scope, activities, and risk areas (Ch. 14.7) (p. 329).
- **Goals and principles.** Validate them, resolve ambiguity, get senior management approval, and outline organization-wide and architecture-specific constraints (timelines, resources) (p. 330).
- **The vision is not a detailed solution.** The Solution Concept Diagram "simply shows the relationships between the organization's goals and objectives on the one hand and the stakeholder requirements and constraints on the other" (p. 332).
- **Value propositions and KPIs.** A business case. Value propositions per stakeholder group for approval. Metrics to monitor whether requirements are met. High-level business risks recorded in the Statement of Architecture Work (p. 333).
- **Business transformation risk.** Change almost always brings risk. Identify the initial level and frequency. Mitigation can wait for the Consolidated Gaps, Solutions, and Dependencies Matrix in Phase E, managed with the Risk Management technique (Ch. 14.8) (pp. 333-334).
- **Closing Phase A.** Secure formal approval of the Statement of Architecture Work plus funding, resources, and time. Communicate the benefits and keep stakeholders engaged (pp. 334-335).

12.4.2 Architecture Vision Phase Artifacts (pp. 336-347)

Artifact (Table 12-7)	Entities used	What it shows	Page
Stakeholder Catalog	None (or stakeholder, once added)	Stakeholders, their influence, key concerns, classification, relevant artifacts	pp. 336-339
Value Chain Diagram	Value stream	High-level primary and support activities, for presentation impact	pp. 339-343
Solution Concept Diagram	Objective, requirement, constraint	A "preliminary pencil sketch" of the solution to align stakeholders	pp. 343-344
Business Model Diagram	Actor, role, stakeholder, contract, function, process, business capability, business service, product	How the organization creates, delivers, and captures value (Business Model Canvas)	pp. 343-345
Business Capability Map	Business capability	Capabilities needed for the goals (high-level at this stage)	pp. 345-346
Value Stream Map	Value stream	End-to-end value-adding activities	pp. 346-347

- **Stakeholder Catalog example (Table 12-8).** CEO Maria Summers (keep satisfied / key player; strategy and goals), CIO Chris West (IT-business alignment), CFO James Noble (keep informed;

budget and cost), HR Manager Anna Carter (minimal effort; workforce adoption). Each is linked to the matrices and catalogs that interest them (pp. 338-339).

- **Why no stakeholder entity?** The Standard cites the sensitivity of stakeholder mapping and Phase A's informal modelling (p. 336).
- **Value Chain Diagram example (Figure 12-4, hospital).** Primary: Patient Admission and Triage; Diagnosis and Treatment; Patient Care and Monitoring; Discharge and Follow-Up. Support: IT and Medical Systems; HR and Training; Facilities and Equipment; Finance and Administration. The colour pairings show which support activity enables which primary one (pp. 340-343).
- **Business Model Diagram (Figure 12-6).** A hospital canvas. It uses entities missing from the foundation metamodel (business collaboration, interaction, interface, resource, value), so consider adding them when tailoring (Ch. 5.2) (p. 345).
- **Capability map gets refined in B.** Phase B adds more capability levels (p. 346).
- **Vision sets scope** for the following phases. Next comes Phase B (p. 347).

Jager's commentary (author's view): Stakeholders defined by role pose little misuse risk, so add a stakeholder entity when tailoring the metamodel (pp. 337-338). The "Business Capability Map" and "Value Stream Map" are really *diagrams* by TOGAF's own definition of map vs. diagram (Ch. 7.4.2), so his examples are drawn as a Business Capability Diagram and a Value Stream Diagram (pp. 346-347).

12.5 Phase B: Business Architecture (pp. 347-398)

- **Why first.** Business Architecture gives multi-dimensional views of capabilities, value delivery, information, and organization, and links them to goals. "It lays the groundwork for all architectural endeavors" and should ideally be the first architecture activity. It also shows stakeholders the business value and return on their support (pp. 347-348).
- **Scope from Phase A.** Strategy states goals, drivers, and metrics but not the path. Business Architecture provides the path (p. 348).

12.5.1 Developing Business Architecture (pp. 348-360)

- **Two objectives.** (1) A Target Business Architecture showing how the organization should function to meet objectives and address the drivers from Phase A. (2) Candidate roadmap components from Baseline-Target gaps. Reuse as much as possible (p. 348).
- **Inputs.** External reference material. Refined principles, goals, and drivers. Assessment of Preliminary capabilities. A Communications Plan. The organizational model (scope, maturity), the RACI, constraints, budget, and the governance and support strategy (p. 349).

Step (Table 12-9)	Description	Output	Page
Select reference models, viewpoints, and tools	Pick repository resources, viewpoints (operations, management, finance), and tools that fit	Views, viewpoints, required building blocks, requirements	pp. 350-352
Develop Baseline and Target Business Architecture description	Describe the current and future state only as far as needed for the vision	Baseline and Target descriptions	pp. 350, 352-353
Perform gap analysis	Check consistency, validate against principles, objectives, and constraints, check completeness, find gaps (Ch. 14.5)	Gap analysis results	pp. 350, 353-354
Define candidate roadmap components	Turn gaps into work packages for a draft roadmap, refined in Phase E	Draft Architecture Roadmap; Business Architecture roadmap components	pp. 350, 354-355

Step (Table 12-9)	Description	Output	Page
Resolve impacts across the Architecture Landscape	Check effects on other architectures and projects, and reuse opportunities	Validated building blocks and artifacts	pp. 351, 355-356
Conduct formal stakeholder review	Executives, process owners, and SMEs revisit the original motivation and the Statement of Architecture Work	Statement of Architecture Work	pp. 351, 356-357
Finalize the Business Architecture	Standardize and document building blocks (Ch. 7.5.2.2), document the rationale for each decision, trace requirements to elements	Documented building blocks and requirements	pp. 351, 357-358
Create the Architecture Definition Document	At minimum Baseline and Target Business Architectures, with visuals, iterated with stakeholder feedback	Architecture Definition Document	pp. 351, 358-359

- **Level of detail.** It depends on how much of the existing structure carries over and whether descriptions already exist. Explore alternative targets with Architecture Alternatives and Trade-Offs (Ch. 14.9) (pp. 352-353).
- **Out-of-scope requirements** go to the Requirements Repository through governed requirements management (Section 12.12) (p. 354).
- **Impact questions.** Does this Business Architecture affect existing architectures? Have recent changes affected it? Can it be reused elsewhere? Does it affect projects, and do projects affect it? (p. 355).
- **Repository use begins.** Artifacts are now recorded in the repository and viewed through the Enterprise Continuum (Ch. 7.3.1). The vision supplies the *why* behind the Statement of Architecture Work (pp. 359-360).

12.5.2 Business Architecture Methods and Modeling Techniques (pp. 360-366)

Technique	What it does	Page
Applying business capabilities	Link the Phase A Business Capability Map to org units, value streams, systems, and objectives	p. 360
Business Capability Heat Mapping	Colour capabilities by maturity, effectiveness, performance, value, or cost. Green = at target, yellow = one level below, red = two or more below, purple = desired but missing (or over-resourced)	pp. 361-362
Applying value streams	Value streams give the *why*, capabilities the *what*. Map stages to capabilities for gap analysis	p. 362
Applying the Organization Map	A network of units, partners, and stakeholders (who owns or uses capabilities and takes part in value streams). Not a reporting chart	pp. 362-363
Applying Information Maps	Map core business information and build matrices from information concepts to capabilities	p. 363
Other modelling	Activity models (BPMN), use-case models, logical data or class models ("is-a" relationships, e.g., sedan, car, vehicle)	pp. 364-365
Business Scenarios	Iteratively test and modify the assumptions behind a project, mainly in Preliminary, A, and B	p. 365

Jager's commentary (author's view): There is "no scientific approach" to rating capability maturity. Heat mapping relies on stakeholder alignment, gut feeling, and common sense (p. 362).

12.5.3 Phase B (Business Architecture) Artifacts (pp. 366-398)

Artifact	Purpose (healthcare example)	Page
Organization/Actor Catalog	Everyone who interacts with IT, including users and owners. Tests requirements for completeness (Board: CEO; Finance: CFO; IT: CIO; Quality and Safety; Clinical Operations)	pp. 369-370
Driver/Goal/Objective Catalog	How drivers become goals and SMART objectives. Reveals synergies between initiatives (e.g., Patient-Centered Care, improve satisfaction, 90% rating by Q4 2028)	pp. 370-371
Role Catalog	Business roles, used for RACI, change impacts, training	pp. 371-372
Business Service/Function Catalog	Filterable functional decomposition. Jager shows it as a Business Service/Function Matrix because cross-mapping is clearer	pp. 372-374
Location Catalog	Physical and virtual locations (campus, data centre, telehealth hub). Supports DR, latency, compliance	pp. 374-375
Process/Event/Control/Product Catalog	Each process with its trigger event, constraining control, and product	pp. 375-376
Contract/Measure Catalog	KPIs, SLAs, OLAs between services and their consumers or providers	pp. 376-377
Business Capabilities Catalog	Capabilities (what, not how) with maturity, owner, and related goal	pp. 377-378
Business Capability Map	Capabilities and sub-capabilities, expanded from Phase A	p. 379
Value Stream (Stages) Catalog / Value Stream Map	Value streams with stages, stakeholder, outcome (Inpatient Care Delivery, Patient Feedback Resolution, Cost Performance Review)	pp. 379-381
Value Stream/Capability Matrix	Links stages to the capabilities that perform them. Key for capability-based planning	pp. 381-382
Strategy/Capability Matrix	Capabilities needed per strategic goal, with priority	p. 382
Capability/Organization Matrix	Which units deliver or contribute to each capability	p. 383
Organization Map / Diagram	Units, partners, suppliers, and their relationships	pp. 383-385
Business Glossary Catalog	Unambiguous business terms. Bridges business and IT. Evolves through the ADM	pp. 385-386
Business Interaction Matrix	Roles vs. functions (performs, supports, informed, not involved)	pp. 386-387
Actor/Role Matrix	Which actors perform which roles (security, skills, training)	pp. 387-388
Business Footprint Diagram	Goals, units, functions, services, and technical components. A CxO communication tool	pp. 388-389
Business Service/Information Diagram	Information each service consumes and produces	pp. 389-390
Functional Decomposition Diagram	Capabilities or functions broken into sub-functions on one page	pp. 390-391
Product Lifecycle Diagram	Products through concept, development, operation, retirement (e.g., Feedback App v2 from Q3 2028)	pp. 391-393
Goal/Objective/Business Service Diagram	Traces goals through objectives to services. Derived from the Footprint Diagram	p. 393
Business Use-Case Diagram	Who (roles) uses which services or processes	p. 394
Organization Decomposition Diagram	Actors, roles, locations in a chain of command	pp. 394-395

Artifact	Purpose (healthcare example)	Page
Process Flow Diagram	Sequential flow with swimlanes, controls, events, products. Good for SME sessions	pp. 395-396
Business Event Diagram	Events triggering processes and outcomes (reactive behaviour)	p. 396
Information Map	Information concepts, categories (transactional = dynamic, event-driven; reference = stable, governed), and relationships	pp. 396-398

- **Phase B feeds the rest.** C structures data and applications around business processes, D aligns technology to business needs, and E identifies projects from business requirements. Give Phase B enough time (p. 398).

Jager's commentary (author's view): The Actor/Role Matrix is one of the least-used artifacts, because HR already holds this information and a second source adds little (p. 387).

12.6 Phase C: Information Systems Architectures (pp. 398-460)

- **Purpose.** A blueprint of the information systems that support the Business Architecture: the systems needed for business capabilities, defined through Data and Application Architectures, with gaps assessed. Key deliverables are data models and application portfolios (p. 398).
- **Two interconnected parts.** Data Architecture (data assets, management, governance: how data is stored, managed, used) and Application Architecture (applications, their interactions with each other and with users, aligned to goals) (p. 399).

Jager's commentary (author's view): BIZBOK's information concept sits between Business and Data Architecture but "is not yet sufficiently reflected" in the Fundamental Content, only in the "Information Mapping" Series Guide (p. 399).

12.6.1 Developing Data Architecture (pp. 400-410)

- **Objectives** mirror Phase B: a Target Data Architecture (entities and relationships, storage, handling, integration) and Data Architecture roadmap components from gaps. Reuse wherever possible (p. 400).
- **Same steps as Phase B (Table 12-31).** Select reference models, viewpoints, and tools; develop Baseline and Target descriptions; perform gap analysis; define candidate roadmap components; resolve impacts across the landscape; conduct a formal stakeholder review; finalize; update the Architecture Definition Document. "The steps don't really differ between the three phases" B, C, and D (pp. 400-402).
- **Differences in emphasis.** Use Data Architecture reference models and viewpoints. Take stock of data assets and flows for the baseline. Explore target options with Alternatives and Trade-Offs (Ch. 14.9). Add data components to the *draft* roadmap from Phase B. Ask whether the Data Architecture affects established architectures and projects, and vice versa (pp. 402-407).
- **Outputs.** An updated Statement of Architecture Work and an updated ADD with Baseline and Target Data Architectures, approved by senior management (pp. 401, 409-410).
- **Aim.** A data infrastructure that meets business goals and is secure and scalable (p. 410).

12.6.2 Phase C (Data Architecture) Artifacts (pp. 410-418)

Artifact (Table 12-32)	Purpose	Page
Data Entity/Data Component Catalog	All data used, with entities related to the data components that structure them. Built from the Information Map	pp. 411-412
Data Entity/Business Function Matrix	Where data is used and who owns it. Supports gap analysis for missing entities	pp. 412-413

Artifact (Table 12-32)	Purpose	Page
Application/Data Matrix	Which applications Create, Read, Update, or Delete which entities (CRUD). Reveals duplication, integration, ownership	p. 414
Conceptual Data Diagram	Relationships between critical entities for business stakeholders. Similar to the Information Map	pp. 414-415
Data Dissemination Diagram	How logical data components are realized by application components. Supports sizing and shows business criticality	p. 415
Data Security Diagram / Catalog	Which roles, units, and applications access which data. Classification and protection (RBAC, encryption, MFA)	pp. 415-416
Data Migration Diagram / Catalog	Extraction from source, transformation and cleansing, loading into target. Overview level in Phase C	pp. 416-417
Data Lifecycle Diagram	Stages of an entity: creation, modification, use, storage, archival, deletion	p. 418

- **Context.** Phase B supplied the business context. Data Architecture influences technology choices (databases, storage, platforms) and runs in parallel with Application Architecture (p. 418).

Jager's commentary (author's view): People know their processes better than their functions, so a **Data Entity/Business Process Matrix** is more useful than the Standard's Data Entity/Business Function Matrix (p. 413).

12.6.3 Developing Application Architecture (pp. 418-427)

- **Objectives** mirror Phase B: a Target Application Architecture, and Application Architecture roadmap components from gaps. Reuse wherever possible (p. 418).
- **Same steps (Table 12-38)** as Phases B and C-Data: select resources and viewpoints, develop Baseline and Target (only what the vision needs, explore options with Alternatives and Trade-Offs), gap analysis, roadmap components added to the Phase B draft, resolve impacts, stakeholder review, finalize, update the ADD (pp. 419-427).
- **Aim.** "A structured, cohesive, and efficient application landscape" whose applications are interoperable and exchange data effectively (p. 427).

12.6.4 Phase C (Application Architecture) Artifacts (pp. 427-441)

Artifact (Table 12-39)	Purpose	Page
Application Portfolio Catalog	All applications with function supported, purpose, lifecycle status, interfaces. Assesses redundancy, alignment, risk	pp. 429-432
Interface Catalog	Interfaces between applications (batch, direct DB, API) to scope dependencies early. Full version adds data, protocol, frequency, security	pp. 432-433
Application/Organization Matrix	Which units use which applications	pp. 433-434
Role/Application Matrix	Which roles use which applications. Essential for role-based access and security	p. 434
Application/Function Matrix	Which applications support which business functions	p. 435
Application Interaction Matrix	Application-to-application communications, with services in the cells (the Interface Catalog as a map)	pp. 435-436
Application Communication Diagram	Data flows between applications across interfaces and protocols	p. 436
Application and User Location Diagram	Where applications are used and hosted, relative to users and locations	p. 437
Application Use-Case Diagram	How actors use application components via services	p. 437

Artifact (Table 12-39)	Purpose	Page
Enterprise Manageability Diagram	How applications are monitored, controlled, supported (monitoring, alerting, logging). A filter of the Communication Diagram	pp. 437-438
Process/Application Realization Diagram	Which applications support or automate each process, with sequencing and integration points	pp. 438-439
Software Engineering Diagram	Breaks applications into packages, modules, services, operations (no example given)	p. 439
Application Migration Diagram	Baseline-to-target application migration, including temporary apps and staging areas, for cost estimation	pp. 439-440
Software Distribution Diagram	Physical applications across physical technology and locations	p. 440

- **Enrich the portfolio.** Add CIA (confidentiality, integrity, availability) scores for security compliance (Figure 12-31), hosting location (on-premises or cloud), and the capabilities automated, shown as coloured views (pp. 431-432).
- **Phase C hand-off.** It bridges business strategy and technology. Its output feeds Phase D, and its gaps drive planning in Phase E (p. 441).

Jager's commentary (author's view): Without an Application Portfolio Catalog, duplicate applications get approved too quickly ("triple or quadruple" functionality) and license costs stay hidden. Rationalization can then take years (pp. 430-431). Adding locations to the portfolio makes the Application and User Location Diagram and the Software Distribution Diagram mostly unnecessary (pp. 437, 440). None of the core roles produces the Software Engineering Diagram (p. 439).

12.7 Phase D: Technology Architecture (pp. 441-459)

- **Purpose.** Define the technology infrastructure (hardware, software, network) and technology standards that support the Business, Data, and Application Architectures, with consistency, scalability, security, interoperability, and future readiness (p. 441).

12.7.1 Developing Technology Architecture (pp. 442-452)

- **Objectives** mirror Phase B: a Target Technology Architecture, and technology roadmap components from gaps. Reuse wherever possible (p. 442).
- **Same steps (Table 12-46)** as B and C. Technology reference models and viewpoints; Baseline and Target (only what the vision needs, with options via Alternatives and Trade-Offs); gap analysis; components added to the Phase B draft roadmap; resolve impacts; stakeholder review (meeting, presentation, feedback, refinement, final approval); finalize building blocks with documented rationale; update the ADD (pp. 443-452).
- **A bridge.** Phase D links conceptual designs to the actual implementation of technology solutions, covering hardware, software, networking, security, and integration (p. 452).

Jager's commentary (author's view): In "nine out of ten organizations" IT has its own separate planning and strategy. TOGAF's idea is that work in *every* domain, technology included, is linked to business goals (p. 442).

12.7.2 Phase D (Technology Architecture) Artifacts (pp. 453-459)

Artifact (Table 12-47)	Purpose	Page
Technology Standards Catalog	Approved technologies, products, versions, and status (e.g., RHEL 10, PostgreSQL 17.5, HL7 FHIR R4 approved / R6 under review)	p. 454
Technology Portfolio Catalog	All technology in use (hardware, infrastructure and application software). The usual starting point of Phase D, fed by registries such as a CMDB	pp. 454-455

Artifact (Table 12-47)	Purpose	Page
Application/Technology Matrix	Which platforms support which applications. Reveals dependencies, reuse, and obsolescence risk	p. 455
Environments and Locations Diagram	Physical and virtual locations where components run. Essentially a filter of the Phase B Location Catalog	p. 456
Platform Decomposition Diagram	Structure of technology platforms: servers, OS, networks, databases grouped into logical and physical platforms	pp. 456-457
Processing Diagram	Components grouped into deployment units and how those connect	p. 457
Networked Computing/Hardware Diagram	Logical application components in a distributed network environment (DMZ, internet, intranet zones). A more detailed Processing Diagram	p. 458
Network and Communications Diagram	How servers, clients, databases, and devices connect across networks: boundaries and paths, not data formats	pp. 458-459

- **Alignment.** Phase D must align with Phase C's applications and is the foundation for Phase E's identification and prioritization of solutions (p. 459).

12.8 Phase E: Opportunities and Solutions (pp. 459-494)

- **Purpose.** Identify and prioritize the projects or programs (work packages) that will implement the Target Architecture. Evaluate buy, build, or reuse. Assess dependencies, opportunities, and constraints to produce a feasible Architecture Roadmap (pp. 459-460).

12.8.1 Defining Opportunities and Solutions (pp. 460-475)

- **Three objectives.** (1) Identify the work packages. (2) Decide whether an incremental approach, with Transition Architectures that each deliver business value, is needed. (3) Define the SBBs needed to complete the target by detailing the ABBs. Phase E produces the *first complete* Architecture Roadmap from the B-D gap analyses and candidate components (p. 460).

Step (Table 12-52)	Description	Output	Page
Determine/confirm key corporate change attributes	Align with the business culture. Assess units' transition capabilities, culture, skills	Implementation Factor Catalog	pp. 461, 463-464
Determine business constraints for implementation	Find drivers that could compromise the target. Mediate with Alternatives and Trade-Offs	n/a	pp. 461, 464-465
Review and consolidate gap analysis results from B-D	One list of gaps, solutions, dependencies. Find SBBs that close several gaps	Consolidated Gaps, Solutions, and Dependencies Matrix	pp. 461, 465-466
Consolidate and reconcile interoperability requirements	Find interoperability conflicts (reused SBBs, COTS, third parties). Either build a translating block or change specifications	n/a	pp. 461, 466-467
Refine and validate dependencies	Group related activities into projects and logical increments	Work Package Portfolio Map	pp. 461, 467-468
Confirm readiness and risk for business transformation	Revisit the Phase A readiness assessment. Classify and mitigate risks	Updated Consolidated Matrix	pp. 461, 468-469
Formulate Implementation and Migration Strategy	Choose the approach and methodology	Implementation and Migration Strategy	pp. 462, 470-471

Step (Table 12-52)	Description	Output	Page
Identify and group major work packages	Group activities to close capability gaps (new, existing, or bought). Classify systems as mainstream, contain, or replace	Work Package Portfolio Map	pp. 462, 471-472
Identify Transition Architecture(s)	Value-delivering milestones fitted to the strategy and capacity for change	Building blocks; Architecture Definition Increments Table; Transition Architectures	pp. 462, 472-473
Create Architecture Roadmap and Implementation and Migration Plan	Consolidate work packages and transitions into a timeline and plan	Architecture Roadmap; Implementation and Migration Plan	pp. 462, 473-475

- **Strategic approaches.** *Greenfield* (completely new), *Revolutionary* (radical overhaul), *Evolutionary* (gradual convergence, parallel running, phased introduction) (p. 470).
- **Implementation methodologies.** *Quick Wins*, *Achievable Targets*, and the *Value Chain Method* (p. 470).
- **Sequencing advice.** Unless there is a good reason otherwise, schedule difficult activities after those that more readily provide missing capabilities. Intervals between transitions need not be equal (p. 472).
- **Implementation and Migration Plan (Table 12-53).** Phases with timeframes, work packages, milestones, and dependencies. Consider a data-driven sequence (applications that create data first) and the lifecycle of existing SBBs (pp. 473-474).
- **Level of detail.** The draft roadmap should match the ADD's level of detail. If much more detail is needed, a different level of architecture may be required (p. 473).
- **Bridge.** Phase E links architecture definition to implementation planning (p. 475).

Jager's commentary (author's view): The Implementation Factor Catalog and the Consolidated Gaps, Solutions, and Dependencies Matrix are not true artifacts, because they contain no metamodel entities (pp. 463-464).

12.8.2 Phase E (Opportunities and Solutions) Artifacts (pp. 475-485)

Artifact (Table 12-54)	Purpose	Page
Project Context Diagram	Scope of work packages: the units, functions, services, processes, apps, data, and technology they affect. Used for PPM and mobilization	p. 477
Benefits Diagram / Catalog	Opportunities classified by size, benefit, complexity (goal, objective, capability, benefit) for prioritization	pp. 477-478
Implementation Factor Catalog	Factor category (risk, assumption, dependency, action, effect), factor, description, and *deduction* (resulting action or constraint). The deductions shape work packages	pp. 478-480
Consolidated Gaps, Solutions, and Dependencies Matrix	Gaps from B-D with domain, potential solutions, dependencies. Links SBBs to ABBs. Clusters similar gaps	pp. 481-483
Architecture Definition Increments Table	Projects and their incremental deliverables across Transition Architectures 1..n	pp. 483-484
Work Package Portfolio Map	Jager's addition (see 12.8.3)	pp. 488-490
Architecture Roadmap	Jager's addition (see 12.8.3)	pp. 490-493

- **Plateau Planning (Figure 12-42).** Baseline, Transition, and Target plateaus, filled with gaps from the consolidated matrix (e.g., the HR recruitment example: new process + Application B in the transition, consolidated information flows in the target). A variant shows projects instead of gaps (Figure 12-43)

(pp. 482-484).

12.8.3 Creating an Architecture Roadmap (pp. 485-494)

- **The roadmap.** A pivotal deliverable built incrementally in E and F from B-D components. It is both a planning and a communication tool, iterative and adaptable, and shows the business value of work packages at each stage (p. 485).
- **Plateaus, gaps, and work packages (Figure 12-44).** Each architecture state is a *plateau*. The *gap* entity sits between plateaus, and *work packages* close the gaps (p. 486).
- **Work packages vs. initiatives vs. projects.** A cluster of work packages is an initiative and often forms a project, program, or portfolio. A work package is done once and produces a defined result (unlike an ongoing process). In agile, it can model a sprint or increment (pp. 486-487).
- **Work package vs. process (Figure 12-45).** Both are series of related tasks producing a result, but a work package is a unique, one-time "process" (p. 487).
- **Work Package Portfolio Map (Table 12-59).** All work packages from strategy or architecture work, each linked to objectives and goals (e.g., WP A "Fit job profile X..." links to the objective "comply with legislation by Q3 2028", which links to the goal "Implement new regulatory requirements"). It is the foundation of the roadmap and lets stakeholders track progress against strategy. Add a column for related projects to avoid duplicate work in Phase F (pp. 488-490).
- **Architecture Roadmap (Figures 12-47, 12-48).** Goals and objectives down the left, work packages plotted over time with start and end dates, status, cost, and owner. It shows the cost of achieving objectives, helps prioritize, and visualizes transitions (pp. 490-493).
- **Never unlinked.** "A roadmap should never be created without linking business goals and objectives to the initiatives." SMART objectives make this measurable (p. 491).
- **Phase E conclusion.** Prioritize and sequence by value, dependencies, and constraints, with stakeholder engagement, ending in a formal agreement on the prioritized projects and solutions (p. 494).

Jager's commentary (author's view): The Standard has a work package entity but no artifact that uses it (other than the Project Context Diagram), so he proposes adding the **Work Package Portfolio Map** and **Architecture Roadmap** as Phase E artifacts (p. 488).

12.9 Phase F: Migration Planning (pp. 495-513)

- **Purpose.** A detailed, step-by-step plan from baseline to target, with Transition Architectures, milestones, resources, and timelines (p. 495).

12.9.1 Creating a Migration Planning (pp. 495-507)

- **Aims.** Prioritize transition projects and activities. Assess dependencies and risks. Plan resources (budget, people, technology). Ensure business continuity (p. 495).
- **Inputs.** The Target Architecture and roadmap components (gaps, opportunities, solutions from B-D, turned into work packages in E), the Implementation Governance Model, and earlier impact and risk insights. The roadmap is *finalized* in Phase F (pp. 495-496).

Step (Table 12-60)	Description	Output	Page
Confirm management framework interactions	Align the plan with Business Planning, EA, Project/Portfolio Management, and Operations Management	Updated Implementation and Migration Plan	pp. 496-499
Assign business value to work packages	Define and measure value. Map work packages to projects. Assign risks from Phase E. Estimate value	Updated Work Package Portfolio Map; Business Value and Risk Matrix	pp. 496, 499-500

Step (Table 12-60)	Description	Output	Page
Estimate resource requirements, project timings, and availability/delivery vehicle	Costs split into capital (establish) and operational (sustain). Savings from decommissioning. Mostly done by project managers, but stay involved	n/a	pp. 496, 501
Prioritize migration projects through cost/benefit assessment and risk validation	Net benefit of SBBs; risk mitigation; stakeholder consensus (architecture team, business, PMs, finance); formal risk review	Risk Management Plan	pp. 496, 501-504
Confirm Architecture Roadmap	Add Transition Architectures, set intervals by value, capability, and risk, consolidate deliverables by project	Transition Architecture State Evolution Table; finalized roadmap	pp. 497, 504-505
Complete Implementation and Migration Plan	Capture external dependencies, resource availability, detailed project plans and milestones	Finalized Implementation and Migration Plan	pp. 497, 505-506
Complete architecture development cycle	Create an Implementation Governance Model if maturity allows. Document lessons learned for Phase H	n/a	pp. 497, 506-507

- **Four management frameworks.** Business Planning (conceives, directs, resources), Enterprise Architecture (structures and contextualizes), Project/Portfolio Management (designs and builds), Operations Management (integrates, operates, maintains). The migration plan may end up inside another framework's plan, with EA participating (pp. 497-498).
- **Business value and risk indices.** Value covers principle adherence, financial contribution, strategic alignment, competitive position. Risk covers size and complexity, technology, organizational capacity, impact of failure. Weights are assigned (p. 500).
- **Risk Management Plan.** At least: document information, approvals, and the approach (tools, recorded information, classification metrics, risk tolerance) (p. 503).
- **Detail check.** The roadmap and plan should match the ADD's detail. Needing much more detail signals a new, more detailed ADM cycle (p. 506).
- **Summary.** Phase F develops the Transition Architectures and a phased, manageable plan that is input to Phase G (p. 507).

12.9.2 Phase F (Migration Planning) Artifacts (pp. 507-513)

Artifact (Table 12-61)	Purpose	Page
Transition Architecture State Evolution Table	Rows of domain plus service, with columns per Transition Architecture describing each service's state (new, in transition, retained, removed)	pp. 508-511
Business Value and Risk Matrix	Projects plotted by business value index (y) against risk index (x), with weighted criteria approved by senior management and colours for at-risk status	pp. 511-513

- **HR example (Figure 12-49).** Transition 1: Application A must be replaced by B. Transition 2: B introduced and services moved from A. Transition 3: B fully operational and A gone. The DBMS persists throughout (it supported A, then B). The cells hold SBBs because they name specific applications (pp. 509-510).
- **Project size** comes from the number of linked work packages, so update the Work Package Portfolio Map with projects (p. 512).

Jager's commentary (author's view): The State Evolution Table is demanding to build per project, but "the pros outweigh the cons" because it details every step per domain and service (p. 511).

12.10 Phase G: Implementation Governance (pp. 513-528)

- **Purpose.** Oversee execution so that the vision "transitions into operational reality". Define structure, roles, and responsibilities. Work with project management to monitor progress and address deviations. "Where the rubber meets the road" (pp. 513, 522).

12.10.1 Establishing Architecture Governance (pp. 514-522)

- **Objectives.** Ensure adherence to the architecture, manage changes during implementation, and continually assess progress. Uses the Governance Framework from Ch. 8 (p. 514).

Step (Table 12-62)	Description	Output	Page
Confirm scope and priorities	Gap analysis to identify needed SBBs (led by Solution Architects). Maximize value from existing investments using the Work Package Portfolio Map. Workshop with development management	Architecture Contract	pp. 514-516
Identify deployment resources and skills	Ensure teams understand EA deliverables and methods, using modelling tools for a shared language. Check team readiness (skills, tools, processes) and dependencies	n/a	pp. 514, 516-517
Guide development of solutions deployment	A project recommendation per implementation: scope (in and out), impact analysis, priorities (Business Value and Risk Matrix), strategic requirements, change requests, functional and non-functional scope. Update the repository	Change requests	pp. 514, 517-518
Perform architecture compliance reviews	Check each building block against standards and principles. Agree success metrics. Get formal sign-off. Create a deployment plan. Track time, cost, quality	Compliance assessments	pp. 515, 519-520
Implement business and IT operations	Deploy service delivery, training, and communication. Publish new baselines to the repository and update the CMDB	Architecture-compliant solutions	pp. 515, 520-521
Perform post-implementation review	Examine results. Raise change requests for deviations (which starts Phase H). Publish reviews and formally close projects	Publication of reviews	pp. 515, 521-522

- **Risk Management throughout G** (Ch. 14.8) (p. 521).
- **Bridge.** G connects designs to real-world deployment, ensuring compliance, change management, and stakeholder engagement (p. 522).

12.10.2 Architecture Performance Monitoring (pp. 522-528)

- **Dashboards and graphs** let all stakeholders monitor progress. There are two areas: work package completion, and goal and objective achievement (pp. 522-523).

12.10.2.1 Measuring progress in completing work packages (pp. 523-525).

Method	How it works	Example (4 activities, 3 complete)
Basic	Equal share per activity (25% each)	75%
Refined	Share proportional to activity duration (12.5%, 25%, 25%, 37.5%)	62.5%, which is more realistic

- Progress can be shown as pie charts per work package (Table 12-63) or as spider or bar charts. The choice is personal preference (p. 525).

12.10.2.2 Measuring achievement of goals and objectives (pp. 526-528).

- **Method.** Average the progress percentages of all work packages linked to a goal or objective via the Architecture Roadmap (p. 526).
- **Example.** Goal A has WP A at 100% and WP B at 75%, so Goal A is 87.5% realized. Goal B has WP C at 50% (on hold) and WP D and E at 0%, so Goal B is 16.7% realized (pp. 526-527, Figures 12-53, 12-54).

- Simple, but useful to both the architecture team and senior management (p. 528).

12.11 Phase H: Architecture Change Management (pp. 528-539)

- **Purpose.** Architecture is not static. Phase H manages change across the architecture's lifecycle: assess impacts, define change procedures, keep the architecture aligned with business goals (p. 528).

12.11.1 Managing Architecture Changes (pp. 528-539)

- **Key objectives.** Continuously assess changes in strategy, business environment, and technology. Have processes and governance to handle change. Evaluate impacts. Implement with minimal disruption. Inform stakeholders. Validate afterward (pp. 528-529).

Step (Table 12-64)	Description	Output	Page
Establish value realization process	Define when change is permitted, the procedures, and what triggers a new ADM cycle. Linked to governance (Ch. 8) and the Architecture Contract	Architecture Compliance Report	pp. 529-531
Deploy monitoring tools	Monitor business and technology change and capability maturity (change management apps, PRINCE2, ITIL, maturity models)	n/a	pp. 529, 532
Manage risks	Identify, classify, mitigate before transformation. Monitor continuously (Ch. 14.8)	Risk Management Plan	pp. 529, 532-533
Provide analysis for architecture change management	Use monitoring data, feedback, and reviews for scenario and impact analyses. Validate with stakeholders. Present as executive summaries or heat maps. Create feedback loops	n/a	pp. 529, 533-534
Develop change requirements to meet performance targets	Use B-D gap results. *Minor* changes go to the framework and principles (update metamodel and Content Framework). *Major* changes need a new Request for Architecture Work	Framework and principle changes; new Request for Architecture Work	pp. 530, 534-536
Manage governance process	Decide who approves and with what criteria. Write policies. Involve the Architecture Board. Use scorecards	Architecture Contract	pp. 530, 536-537
Activate the process to implement change	Phased rollout plan with timelines, dependencies, milestones. Communicate. Monitor and refine	n/a	pp. 530, 537-538

- **Architecture Compliance Report.** States whether a change complies. Non-compliance may get an exemption with valid rationale. High-impact changes need a strategy to manage them (p. 531).
- **Criteria mature over time.** Organizations' risk appetites differ, so each ADM iteration moves governance toward tailored criteria and higher maturity (Ch. 15) (p. 531).
- **Why it matters.** "Too often, organizations are left with an Enterprise Architecture that fit yesterday's organization." Change management lets the organization adapt quickly, reduce risk, and keep alignment with business goals (pp. 535, 539).
- **In short.** Identify drivers for change, formulate proposals (rationale, benefits, impact), analyze risk, cost, and impact, govern approval, plan, implement with coordination, then monitor and gather feedback (pp. 538-539).

12.12 Requirements Management (pp. 539-547)

- **Role.** Gather and document stakeholder requirements, and make them available to other phases. Track, categorize, prioritize, and version them. Refine them continuously, check them against the vision, and share them for transparency (pp. 539-540).

12.12.1 Capture, Manage, and Refine Requirements (pp. 540-545)

- **Not a separate phase.** A continuous activity throughout the ADM. Its key objective is that the requirements management process is available and applied in *all* phases (p. 540).

#	Requirements Management step	ADM phase step
1	n/a	Identify and document requirements (Phase A, recorded in the Requirements Repository)
2	Establish baseline requirements in the Requirements Repository (from B, C, D, and E)	n/a
3	Monitor baseline requirements (keep them relevant as business, regulation, and technology change)	n/a
4	n/a	Identify new and changed requirements (as priorities shift)
5	Identify changed requirements and record priorities (validate with stakeholders, manage conflicts)	n/a
6	n/a	Assess impact of changed requirements on current and previous phases (implement now or in a later iteration)
7	n/a	Implement requirements arising from Phase H (make them available for future iterations)
8	Update the Architecture Requirements Repository	n/a
9	n/a	Implement change in the current phase (no Requirements Management action)
10	n/a	Assess and revise gap analysis for past phases (record "gap requirements" and revise the target)

(Table 12-65, pp. 541-545)

12.12.2 Requirements Management Artifacts (pp. 545-547)

- **Requirements Catalog** (entities: requirement, objective, goal, assumption, constraint, gap). It captures what the organization must do to meet its objectives. Link each requirement to goals or objectives for traceability. Requirements are implemented through change initiatives scoped in Phase E and also serve as a quality check that an architecture is fit for purpose. It is often accompanied by a requirements management plan (pp. 545-546).
- Example: "A centralized digital feedback platform..." links to *Improve patient satisfaction* and *Enhance patient safety* (Table 12-67, p. 546).

12.13 The Architecture Development Method in Short (pp. 547-557)

- **A logical method.** It groups key activities to clarify relationships and information flow, with steps that *can* be performed (p. 547).
- **Phase-by-phase recap** of all steps for the Preliminary Phase, A, B-D, E, F, G, H, and Requirements Management (pp. 548-556), summarized as a poster in Figure 12-56 (p. 557).

Jager's commentary (author's view): The most common complaint about TOGAF, that "everything in the ADM has to be implemented", is "completely wrong and misguided". The ADM prescribes neither that all steps be done nor their order. It is iterative (Ch. 13.3), and skipping steps (e.g., in agile) "is actually supported and even encouraged" (pp. 547-548).

12.14 Summary (pp. 557-560)

- The ADM is a logical method, not a process. It is the heart of the Standard, comprehensive and iterative (p. 557).
- It has ten phases with distinct objectives (repeated from 12.1) (pp. 558-559).
- It describes what *can* be done, does not prescribe order, and must be tailored. Every phase and artifact has been illustrated (pp. 559-560).

Key Terms & Definitions

Term	Definition	Page
Architecture Development Method (ADM)	TOGAF's iterative, tailorable method of ten phases for developing and managing EA	pp. 295-296
Request for Architecture Work	Sponsor's formal request that triggers an ADM cycle	pp. 300, 302
Statement of Architecture Work	Scope and approach of the cycle. Effectively a contract between architects and sponsor	pp. 301, 303
Architecture Definition Document	Deliverable container for core artifacts across domains and states	pp. 301, 303
Architecture Contract	Joint agreement on deliverables, quality, and fitness for purpose. Starts implementation governance	pp. 301, 304
Baseline / Transition / Target Architecture	Current state / intermediate value-delivering state / desired future state	pp. 298, 472
Gap analysis	Comparison of baseline and target to find what is missing, to reuse, or to remove	pp. 304, 353
Plateau	A stable architecture state (baseline, transition, target) on the roadmap	p. 486
Work package	A set of actions to achieve one or more business objectives. Part of a project, a project, or a program	p. 487
Architecture Roadmap	Work packages plotted over time against goals and objectives, from baseline to target	pp. 485, 490-493
Implementation and Migration Plan	Activities, timeframes, milestones, and dependencies to realize the roadmap	pp. 473-474, 505
Business Transformation Readiness Assessment	Evaluation of readiness factors (vision, leadership, engagement, governance, culture)	pp. 329, 468
Greenfield / Revolutionary / Evolutionary	New build / radical overhaul / gradual convergence implementation approaches	p. 470
Architecture Compliance Report	States whether a change complies with the current architecture	p. 531
Heat mapping	Colour-coding capabilities or value stream stages by maturity, performance, value, or cost	pp. 361-362

Figures & Tables Referenced

Figure / Table	What it shows	Page
Figure 12-1	The Architecture Development Method	p. 297
Figure 12-2	Document creation during the ADM	p. 302
Tables 12-1, 12-2, 12-3	Preliminary Phase steps. Example EA team RACI	pp. 306, 314, 315

Figure / Table	What it shows	Page
Figure 12-3	Driver/Goal/Objective Diagram (hospital)	p. 309
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Table 12-9, Figure 12-9	Phase B steps. Business Capability Heat Map	pp. 350, 361
Tables 12-10 to 12-30, Figures 12-10 to 12-25	Phase B artifacts	pp. 367-398
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Tables 12-52 to 12-59, Figures 12-41 to 12-48	Phase E steps, artifacts, Plateau Planning, Work Package Portfolio Map, Architecture Roadmap	pp. 461-493
Tables 12-60, 12-61, Figures 12-49, 12-50	Phase F steps. State Evolution Table. Business Value and Risk Matrix	pp. 496-512
Tables 12-62, 12-63, Figures 12-51 to 12-54	Phase G steps. Progress measurement methods	pp. 514-527
Table 12-64	Phase H steps	pp. 529-530
Tables 12-65 to 12-67, Figure 12-55	Requirements Management steps, artifacts, catalog	pp. 541-546
Figure 12-56	Summary of all ADM steps (poster)	p. 557

Self-Check Questions

Q: Is the ADM a prescriptive, sequential process?

A: No. It describes everything one *can* do, not what must be done. It does not fix the order, it is iterative, and it must be tailored. It only looks like a waterfall (pp. 295-296, 547-548).

Q: What are the two objectives of the Preliminary Phase?

A: To determine the Architecture Capability the organization needs, and to establish it (pp. 297, 305).

Q: Name the four key ADM documents and when each is created.

A: Request for Architecture Work (Preliminary); Statement of Architecture Work (Phase A); Architecture Definition Document (built in B-D, approved in E, finalized in F); Architecture Contract (from E, governing implementation) (pp. 300-305).

Q: What two objectives do Phases B, C, and D share?

A: Craft the Target Architecture for the domain, and identify candidate Architecture Roadmap components from Baseline-Target gaps (pp. 298, 348).

Q: What does the Solution Concept Diagram actually show?

A: Not a detailed solution, but the relationships between the organization's goals and objectives and the stakeholders' requirements and constraints, as an early "pencil sketch" (pp. 332, 343).

Q: Explain Business Capability Heat Mapping colours in Jager's maturity example.

A: Green is at the desired level. Yellow is one level below. Red is two or more levels below (or not functioning). Purple is not yet in place but desired (pp. 361-362).

Q: What are the three objectives of Phase E?

A: Identify the work packages, decide whether an incremental approach (Transition Architectures) is needed, and define the building blocks (SBBs) needed to complete the target (pp. 298, 460).

Q: Name the three strategic implementation approaches and three methodologies in Phase E.

A: Approaches: Greenfield, Revolutionary, Evolutionary. Methodologies: Quick Wins, Achievable Targets, Value Chain Method (p. 470).

Q: Why does Jager add a Work Package Portfolio Map and Architecture Roadmap as Phase E artifacts?

A: The Standard has a work package entity but barely uses it. These artifacts link every initiative to goals and objectives, so strategy realization can be planned and tracked (pp. 488-491).

Q: Which four management frameworks must the Implementation and Migration Plan align with in Phase F?

A: Business Planning, Enterprise Architecture, Project/Portfolio Management, Operations Management (pp. 497-498).

Q: In Phase G, how does the refined method of measuring work package progress differ from the basic one?

A: The basic method gives each activity an equal share. The refined method weights by duration. With three of four activities done, 75% (basic) versus 62.5% (refined, more realistic) (pp. 523-524).

Q: How is goal realization progress calculated?

A: Average the progress of the work packages linked to the goal. E.g., WP A at 100% and WP B at 75% gives 87.5% for Goal A (pp. 526-527).

Q: In Phase H, what distinguishes a minor from a major change?

A: Minor changes update the framework or principles (and so the metamodel and Content Framework). Major changes require a new Request for Architecture Work and a new ADM cycle (p. 535).

Q: Why is Requirements Management at the centre of the ADM rather than a phase?

A: It is a continuous activity that captures, manages, and refines requirements and makes them available to every phase throughout the cycle (pp. 299, 540).

Q: Which Phase B artifact would you use to show which capabilities each value stream stage needs?

A: The Value Stream/Capability Matrix (or Diagram), a key tool for capability-based planning (pp. 381-382).